

NAME

calibrate -- Apply extinction corrections and flux calibrations

USAGE

calibrate input output [records]

PARAMETERS**input**

List of input spectra to be calibrated. When using record format extensions the root names are specified, otherwise full image names are used.

output

List of calibrated spectra. If no output list is specified or if output name is the same as the input name then the calibrated spectra replace the input spectra. When using record format extensions the output names consist of root names to which the appropriate record number extension is added. The record number extension will be the same as the input record number extension. The output spectra are coerced to have real datatype pixels regardless of the pixel type.

records(imred.irsand imred.iidsonly)

The set of record number extensions to be applied to each input and output root name when using record number extension format. The syntax consists of comma separated numbers or ranges of numbers. A range consists of two numbers separated by a hyphen. This parameter is not queried when record number formats are not used.

extinct= yes

Apply extinction correction if a spectrum has not been previously corrected? When applying an extinction correction, an extinction file is required.

flux = yes

Apply a flux calibration if a spectrum has not been previously calibrated? When applying a flux calibration, sensitivity spectra are required.

extinction= <no default>

Extinction file for the observation. Standard extinction files are available in the "onedstds\$" directory.

observatory =")_observatory"

Observatory at which the spectra were obtained if not specified in the image header by the keyword OBSERVAT. The default is a redirection to the package parameter of the same name. The observatory may be one of the observatories in the observatory database, "observatory" to select the observatory defined by the environment variable "observatory" or the parameter **observatory.observatory**, or "obsvars" to select the current parameters in the **observatory** task. See **observatory** for additional information.

ignoreaps= no

Ignore aperture numbers and apply a single flux calibration to all apertures? Normally multiaperture instruments have separate sensitivity functions for each aperture while long slit or Fabry-Perot data use a single sensitivity function where the apertures are to be ignored. The sensitivity spectra are obtained by adding the aperture number as an extension to the sensitivity spectrum root name. When apertures are ignored the specified sensitivity spectrum name is used without adding an extension and applied to all input apertures.

sensitivity = "sens"

The root name for the sensitivity spectra produced by **sensfunc**. Normally with multiaperture instruments, **sensfunc** will produce a spectrum appropriate to each aperture with an aperture number extension. If the apertures are ignored (*ignoreaps* = yes) then the sensitivity spectrum specified is used for all apertures and no aperture number is appended automatically.

fnu = no

The default calibration is into units of flux per unit wavelength (F-lambda). If *fnu* = yes then the calibrated spectrum will be in units of flux per unit frequency (F-nu).

airmass,exptime

If the airmass and exposure time are not in the header nor can they be determined from other keywords in the header then these query parameters are used to request the airmass and exposure time. The values are updated in the input and output images.

DESCRIPTION

The input spectra are corrected for extinction and calibrated to a scale using sensitivity spectra produced by the task **sensfunc**. One or both calibrations may be performed by selecting the appropriate parameter flags. It is an error if no calibration is specified. Normally the spectra should be extinction corrected if also flux calibrating. The image header keywords DC-FLAG (or the dispersion type field in the "multispec" world coordinate system), EX-FLAG, and CA-FLAG are checked for dispersion solution (required), previous extinction correction, and previous flux calibration. If previously calibrated the spectrum is skipped and a new output image is not created.

The input spectra are specified by a list of root names (when using record extension format) or full image names. The output calibrated spectra may replace the input spectra if no output spectra list is specified or if output name is the same as the input name. When using record number extensions the output spectra will have the same extensions applied to the root names as those used for the input spectra.

When applying an extinction correction the AIRMASS keyword is sought. If the keyword is not present then the airmass at the time defined by the other header keywords is computed using the latitude of the observatory and observation parameters in the image header. The observatory is first determined from the image under the keyword OBSERVAT. If absent the observatory specified by the task parameter "observatory" is used. See **observatory** for further details of the observatory database. If the air mass cannot be determined an error results. Currently a single airmass is used and no correction for changing extinction during the observation is made and adjustment to the middle of the exposure. The task **setairmass** provides a correction for the exposure time to compute an effective air mass. Running this task before calibration is recommended.

If the airmass is not in the header and cannot be computed then the user is queried for a value. The value entered is then recorded in both the input and output image headers. Also if the exposure time is not found then it is also queried and recorded in the image headers.

The extinction correction is given by the factor

$$10. ** (0.4 * \text{airmass} * \text{extinction})$$

where the extinction is the value interpolated from the specified extinction file for the wavelength of each pixel. After extinction correction the EX-FLAG is set to 0.

When applying a flux calibration the spectra are divided by the aperture sensitivity which is represented by a spectrum produced by the task **sensfunc**. The sensitivity spectrum is in units of:

$$2.5 * \text{Log}_{10} [\text{counts/sec/Ang} /$$

A new spectrum is created in "F-lambda" units - or "F-nu" units - ergs/cm2/sec/Hz. The sensitivity must span the range of wavelengths in the spectrum and interpolation is used if the wavelength coordinates are not identical. If some pixels in the spectrum being calibrated fall outside the wavelength range of the sensitivity function spectrum a warning message giving the number of pixels outside the range. In this case the sensitivity value for the nearest wavelength in the sensitivity function is used.

Multiaperture instruments typically have a separate aperture sensitivity function for each aperture. The appropriate sensitivity function for each input spectrum is selected based on the spectrum's aperture by appending this number to the root sensitivity function spectrum name. If the *ignoreaps* flag is set, however, the aperture number relation is ignored and the single sensitivity spectrum (without extension) is applied.

EXAMPLES

1. To flux calibrates a series of spectra replacing the input spectra by the calibrated spectra:

```
cl> calibrate nite1 ""
```

2. To only extinction correct echelle spectra:

```
cl> calibrate ccd*.ec.imh new//ccd*.ec.imh flux-
```

3. To flux calibrate a long slit spectrum:

```
cl> dispaxis = 2
cl> calibrate obj.imh fcobj.imh
```

REVISIONS

CALIBRATEV2.10.3

This task was revised to operate on 2D and 3D spatial spectra; i.e. long slit and Fabry-Perot data cubes. This task now includes the functionality previously found in **longslit.extinction** and **longslit.fluxcalib**.

A query for the airmass and exposure time is now made if the information is not in the header and cannot be computed from other header keywords.

CALIBRATEV2.10

This task was revised to operate on nonlinear dispersion corrected spectra and 3D images (the **apextract** "extras"). The aperture selection parameter was eliminated (since the header structure does not allow mixing calibrated and uncalibrated spectra) and the latitude parameter was replaced by the observatory parameter. The observatory mechanism insures that if the observatory latitude is needed for computing an airmass and the observatory is specified in the image header the correct calibration will be applied. The record format syntax is available in the **irs/iids** packages. The output spectra are coerced to have real pixel datatype.

SEE ALSO

setairmass, standard, sensfunc, observatory, continuum